



Predictors of pharmacoresistant epilepsy

Nikolas Hitiris^a, Rajiv Mohanraj^b, John Norrie^c,
Graeme J. Sills^a, Martin J. Brodie^{a,*}

^a *Epilepsy Unit, Division of Cardiovascular and Medical Sciences, Western Infirmary, Glasgow G11 6NT, Scotland, UK*

^b *Department of Neurology, Hope Hospital, Salford, UK*

^c *Centre for Healthcare Randomised Trials (CHaRT), Health Services Research Unit, University of Aberdeen, Scotland, UK*

Received 16 March 2007; received in revised form 24 May 2007; accepted 2 June 2007

KEYWORDS

Pharmacoresistant
epilepsy;
Predictive factors;
Seizures;
Psychiatric
comorbidity

Summary Outcome data were analysed from 780 patients newly diagnosed with epilepsy and followed up at a single centre over a 20-year period to investigate which clinical factors predicted pharmacoresistance. Patients were divided at the time of analysis into those whose seizures had been controlled for at least the last 12 months of follow up ($n=462$) and those whose epilepsy remained refractory ($n=318$). Numbers of pre-treatment seizures were greater in uncontrolled patients. Those reporting more than 10 seizures prior to initiation of therapy were more than twice as likely to develop refractory epilepsy. Univariate and multivariate logistic regression analyses demonstrated that pharmacoresistance was also associated with family history of epilepsy, previous febrile seizures, traumatic brain injury as the cause of the epilepsy, intermittent recreational drug use, and prior or current psychiatric comorbidity, particularly depression. Factors not predicting poorer outcome included gender, neurological deficit and mental retardation. The most interesting new finding was the correlation between psychiatric comorbidity and lack of response to antiepileptic drug therapy. The deleterious neurobiological processes that underpin depression, anxiety and psychosis may interact with those producing seizures to increase the extent of brain dysfunction and thereby the likelihood of developing pharmacoresistant epilepsy.

© 2007 Elsevier B.V. All rights reserved.

Introduction

Epilepsy, the commonest serious neurological disorder, typically responds well to antiepileptic drug (AED) therapy.

However, 30–40% of adult patients remain refractory to pharmacological treatment (Annegers et al., 1979; Cockerell et al., 1995; Kwan and Brodie, 2000). Uncontrolled epilepsy is associated with cognitive deterioration, psychosocial dysfunction, and increased morbidity and mortality (Devinsky, 1999). If refractory epilepsy could be recognised early, more aggressive pharmacotherapy or early surgery, where appropriate, could potentially ameliorate these adverse consequences.

* Corresponding author. Tel.: +44 141 211 2944;
fax: +44 141 334 9329.

E-mail address: mjb2k@clinmed.gla.ac.uk (M.J. Brodie).

A diverse range of clinical predictors has been mooted for refractory epilepsy, including abnormal neurological examination, electroencephalography and brain imaging (Elwes et al., 1984; Hauser et al., 1990; Sillanpää, 1993; Mattson et al., 1996; Berg et al., 1996, 2001; Cockerell et al., 1997; Semah et al., 1998; Ko and Holmes, 1999). A large number of pre-treatment seizures is also known to be a poor prognostic indicator (Collaborative Group for the Study of Epilepsy, 1992; Sillanpää, 1993; Kwan and Brodie, 2000; Leschziner et al., 2006). We investigated which clinical factors helped to predict pharmacoresistance in a cohort of patients with newly diagnosed epilepsy followed over a 20-year period in a single centre. Only information that was available prior to treatment introduction was considered in this analysis.

Methods

A total of 780 patients living in the West of Scotland diagnosed with epilepsy and prescribed their first AED in the Epilepsy Unit of the Western Infirmary in Glasgow, Scotland, between July 1982 and May 2001 were included in this analysis. Outcomes were followed until the end of December 2003. No patients had previously received treatment with an AED for any indication. A structured protocol was employed at the initial consultation in which demographic and clinical parameters were recorded. All patients received monotherapy initially and treatment regimens were altered as necessary depending on efficacy and tolerability.

Outcomes were obtained from case notes and seizure diaries. Response to treatment was defined as achieving seizure freedom for at least the last 12 months of follow-up. The remaining patients were labelled as having refractory epilepsy. Patients who did not return for review or were persistently non-adherent to their treatment regimen were excluded from the analysis. Prognosis in relation to later syndrome classification and response to sequential AED regimens has been published elsewhere (Mohanraj and Brodie, 2005, 2006, 2007). No patient underwent epilepsy surgery nor developed seizures after AED withdrawal.

The following information was obtained prior to diagnosis and initiation of treatment: family history of epilepsy, history of febrile seizures, potential aetiological factors, intermittent recreational drug use and prior or current psychiatric comorbidity. A positive family history was recorded when epilepsy was present in at least one first-degree relative. Recreational drugs included cocaine, amphetamines and cannabis. None of the patients included in the study regarded themselves or were documented by their general practitioners as having a "drugs problem". Patients whose epilepsy was a consequence of traumatic brain injury were identified following the criteria outlined by Annegers et al. (1998). Psychiatric comorbidity consisted of depression, anxiety, or psychosis, which preceded or accompanied the diagnosis of epilepsy. This information was obtained from the patient's history and no systematic assessment of psychiatric comorbidity was undertaken.

The Mann-Whitney test was employed to compare baseline (at study entry) characteristics between patients with controlled and

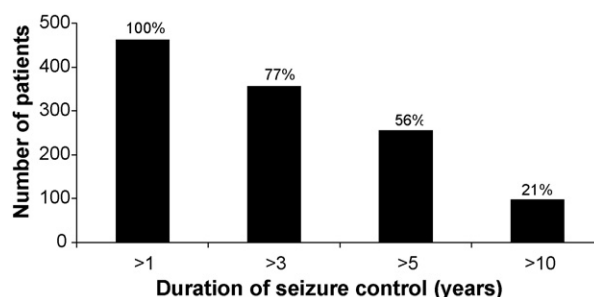


Figure 1 Duration of seizure control in responders to antiepileptic drug therapy.

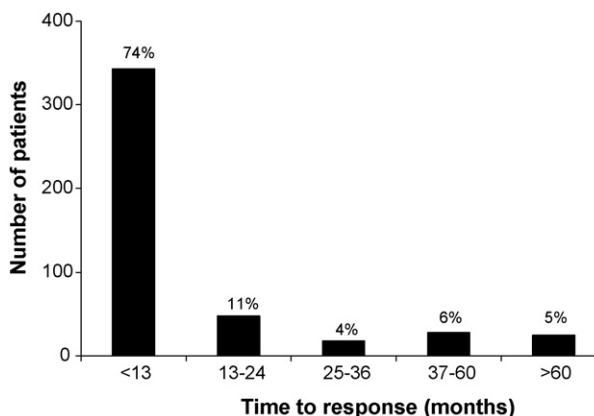


Figure 2 Time to response following initiation of antiepileptic drug therapy.

refractory epilepsy for continuous data. Logistic regression on the outcome "seizure-free for the last 12 months? Yes/No" was used to investigate covariates of interest, first individually in univariate models and then together in a multivariate logistic regression. All analyses were undertaken with Minitab for Windows (version 13.32). $P < 0.05$ was considered as indicating statistical significance. No correction was made for multiple testing.

Results

There were 462 (59.2%) patients with controlled epilepsy at the time of analysis, most of whom had been seizure-free for more than 5 years. (Fig. 1). Time to treatment response, i.e., seizure freedom for at least the previous 12 months, is outlined in Fig. 2. The vast majority of responders could be identified shortly after starting AED therapy. Median age at the time of diagnosis was similar between the groups (Table 1). Numbers of pre-treatment seizures were greater and duration of pre-treatment epilepsy was

Table 1 Demographic characteristics in patients with controlled and refractory epilepsy

Demographic factor	Controlled (n = 462)	Refractory (n = 318)	P-value
Male/female	228/234	177/141	NS
Median age at diagnosis (years)	31 (range 9–93)	31 (range 13–81)	NS
Median duration of pre-treatment seizures (years)	0.67 (range 0.1–57)	1 (range 0.1–44)	<0.0001
Median number of pre-treatment seizures	4 (range 1 to >100)	7 (range 1 to >100)	<0.0001
Median number of seizures 3 months pre-treatment	2 (range 1 to >100)	3 (range 1 to >100)	<0.0003

Table 2 Univariate and multivariate logistic regression analysis for predictors of refractory epilepsy

	Controlled	Refractory	Univariate			Multivariate		
			Odds ratio	95% Confidence intervals	P-value	Odds ratio	% Confidence intervals	P-value
Male gender	228	177	1.29	0.97–1.72	0.083	1.13	0.82–1.55	0.45
Family history	40	51	2.02	1.3–3.13	0.002	1.89	1.15–3.00	0.011
Febrile seizures	12	23	2.92	1.43–5.97	0.003	3.36	1.58–7.18	0.002
Traumatic brain injury	25	50	3.26	1.97–5.40	<0.001	2.73	1.59–4.69	<0.001
Psychiatric comorbidity	34	50	2.35	1.48–3.73	<0.001	2.17	1.33–3.55	0.002
Neurological deficit	19	15	1.15	0.58–2.31	0.685	1.47	0.70–3.08	0.31
Mental retardation	16	12	1.03	0.48–2.18	0.946	1.00	0.45–2.24	0.99
Recreational drug use	9	30	4.76	2.37–9.58	<0.001	4.26	2.03–8.94	<0.001
More than 10 seizures before treatment	105	141	2.71	1.99–3.69	<0.001	2.77	1.98–3.89	<0.001
Seizures for >1 year before treatment	193	158	1.38	1.03–1.83	0.029	1.06	0.71–1.47	0.72

longer in patients with refractory epilepsy (Table 1). Those experiencing more than 10 seizures prior to initiation of treatment were more than twice as likely to develop refractory epilepsy (Table 2).

Univariate logistic regression comparing patients with controlled and refractory epilepsy demonstrated that pharmacoresistance was also associated with family history of epilepsy, prior febrile seizures, traumatic brain injury as a cause of the epilepsy, intermittent recreational drug use and psychiatric comorbidity (Table 2). Depression alone accounted for 85% of the psychiatric diagnoses and was itself associated with lack of response to treatment (odds ratio 2.27; 95% confidence intervals 1.38–3.73; $P=0.001$). Most (80%) of these patients had symptomatic or cryptogenic epilepsy. Gender, neurological deficit and the presence of mental retardation did not predict a poorer outcome.

A multivariable logistic regression model demonstrated that family history of epilepsy, febrile seizures, traumatic brain injury, recreational drug use, psychiatric comorbidity and more than 10 pre-treatment seizures all remained significantly associated with a worse prognosis (Table 2). Duration of pre-treatment epilepsy did not remain statistically significant in the multivariate analysis.

Discussion

A simple definition of treatment failure: "seizure-free for at least the last 12 months? Yes/No" was chosen and logistic regression used to model the influence of covariates of interest in exploring predictors of "refractoriness". Patients had been in the study for a minimum of 2.5 and a maximum of 21 years at the census date of December 2003, and so had different lengths of follow up, and hence different opportunities to remain seizure-free for at least the previous 12 months. With such data an alternative approach would have been a time-to-event analysis, such as a Cox proportional hazards regression. However, the simplicity of a logistic regression was preferred since 74% of patients with controlled epilepsy achieved 12 months' seizure freedom shortly after starting treatment (Mohanraj and Brodie,

2006). A survival model approach would have been expected to yield similar results. No adjustments were made for multiple statistical testing given the number of possible approaches that could be taken using a range of disparate methodologies. Care must be taken, therefore, not to over-interpret the data.

This study supports the observation of worse outcomes with increased pre-treatment seizure frequency (Collaborative Group for the Study of Epilepsy, 1992; Sillanpää, 1993; MacDonald et al., 2000; Kwan and Brodie, 2000; Leschziner et al., 2006). This was particularly the case in those patients reporting a greater number of events immediately prior to diagnosis and initiation of therapy. Those experiencing more than 10 seizures before starting on their first AED were more than twice as likely to develop refractory epilepsy.

Retrospective studies have estimated that 10–15% of patients with epilepsy had previous febrile convulsions (Camfield et al., 1994; Hamati-Haddad and Abou-Khalil, 1998). These events were usually associated with later temporal lobe epilepsy, but could also be antecedents for other syndromes (Hamati-Haddad and Abou-Khalil, 1998). Factors predicting the development of epilepsy following febrile seizures included family history, recurrent and complex episodes, and neurodevelopmental abnormalities (Nelson and Ellenberg, 1976; Annegers et al., 1987; Berg and Shinnar, 1996). Twin studies have suggested that genes and, therefore, family history may also be relevant to the pathogenesis of epilepsy (Berkovic et al., 1998; Kjeldsen et al., 2003), but less so to the prognosis (Johnson et al., 2003). Febrile seizures and family history both appeared to be risk factors for refractory epilepsy in our adult population.

Traumatic brain injury accounts for around 3% of people with epilepsy (Sander et al., 1990). Work by Annegers and colleagues demonstrated that there was a significant increased risk of developing epilepsy with severe trauma and a smaller chance with mild to moderate head injuries (Annegers et al., 1998). This study reports an association between treatment resistant epilepsy and previous traumatic brain injury, perhaps as a consequence of the diffuseness of the cerebral damage in these individuals.

The finding that intermittent recreational drug use was likely to predict pharmacoresistance remains unexplained. This is a difficult covariate to measure accurately. In this study, we relied on self-reporting which, given the sensitive nature of the activity, may be subject to underestimation. This behaviour may also be a marker of unrecognised incomplete compliance with treatment. Another possibility is that intermittent use of these substances may be direct provoking factors in reducing seizure threshold (Smith and McBride, 1999). If confirmed by a more rigorous study, this observation has substantial relevance to the everyday management of epilepsy given the high frequency of this behaviour in the general population (Zagnoni and Albano, 2002).

Psychiatric disorders, particularly depression, are common in people with epilepsy (Mendez et al., 1986; Kanner, 2003; Gilliam et al., 2006). Furthermore, a history of depression and attempted suicide increases the risk of later unprovoked seizures (Forsgren and Nystrom, 1990; Hesdorffer et al., 2000, 2006). This study suggests that a history of psychiatric comorbidity is also associated with treatment-resistant epilepsy. However, data on psychiatric comorbidities were not collected systemically using validated instruments and so this observation requires confirmation preferably in a prospective study in newly diagnosed epilepsy.

Some antidepressants can lower seizure threshold (Pisani et al., 2002). However, possible temporal lobe structural and functional abnormalities and abnormal brain secretion of neurotransmitters, such as serotonin, are among some of the common pathogenetic mechanisms linking temporal lobe epilepsy, in particular, with major depression (Kanner, 2006). Our observation of poor pharmacological outcomes in patients with comorbid depression is supported by that of Kanner et al. (2006) who recently reported similar findings in patients undergoing temporal lobectomy. A lifetime history of depression was the sole predictor of post-surgical auras in seizure-free patients and also a predictor of disabling seizures in those with a poorer outcome. Thus, the deleterious neurobiological processes that underpin depression, anxiety and psychosis may interact with those producing seizures to increase the extent of brain dysfunction and thereby the likelihood of developing pharmacoresistant epilepsy (Jobe, 2003).

In conclusion, a number of factors available at the time of diagnosis and initiation of treatment were predictive of subsequent pharmacoresistance. These included increased number of pre-treatment seizures, family history, prior febrile convulsions, traumatic brain injury as a cause of the epilepsy, intermittent recreational drug use and previous or current psychiatric comorbidity. Most of these variables increased the odds of developing refractory epilepsy two to threefold. Factors not predicting poorer outcome included gender, neurological deficit and the presence of mental retardation.

References

- Annegers, J.F., Hauser, W.A., Elveback, L.R., 1979. Remission of seizures and relapse in patients with epilepsy. *Epilepsia* 20, 729–737.
- Annegers, J.F., Hauser, W.A., Coan, S.P., Rocca, W.A., 1998. A population-based study of seizures after traumatic brain injury. *N. Engl. J. Med.* 338, 20–24.
- Berg, A.T., Shinnar, S., 1996. Unprovoked seizures in children with febrile seizures: short-term outcome. *Neurology* 47, 562–568.
- Berg, A.T., Levy, S.R., Novotny, E.J., Shinnar, S., 1996. Predictors of intractable epilepsy in childhood: a case-control study. *Epilepsia* 37, 24–30.
- Berg, A.T., Shinnar, S., Levy, S.R., Testa, F.M., Smith-Rapaport, S., Beckerman, B., 2001. Early development of intractable epilepsy in children: a prospective study. *Neurology* 56, 1445–1452.
- Berkovic, S.F., Howell, R.A., Day, D.A., Hopper, J.L., 1998. Epilepsies in twins: genetics of the major epilepsy syndromes. *Ann. Neurol.* 43, 435–445.
- Camfield, P., Camfield, C., Gordon, K., Dooley, J., 1994. What types of epilepsy are preceded by febrile seizures? A population-based study of children. *Dev. Med. Child Neurol.* 36, 887–892.
- Collaborative Group for the Study of Epilepsy 1992. Prognosis of epilepsy in newly referred patients: a multicenter prospective study of the effects of monotherapy on the long-term course of epilepsy. *Epilepsia* 33, 45–51.
- Cockerell, O.C., Johnson, A.L., Sander, J.W., Hart, Y.M., Shorvon, S.D., 1995. Remission of epilepsy: results from the national general practice study of epilepsy. *Lancet* 346, 140–144.
- Cockerell, O.C., Johnson, A.L., Sander, J.W., Shorvon, S.D., 1997. Prognosis of epilepsy: a review and further analysis of the first nine years of the British National General Practice Study of Epilepsy, a prospective population-based study. *Epilepsia* 38, 31–46.
- Devinsky, O., 1999. Patients with refractory seizures. *N. Engl. J. Med.* 340, 1565–1570.
- Elwes, R.D., Johnson, A.L., Shorvon, S.D., Reynolds, E.H., 1984. The prognosis for seizure control in newly diagnosed epilepsy. *N. Engl. J. Med.* 311, 944–947.
- Forsgren, L., Nystrom, L., 1990. An incident case-referent study of epileptic seizures in adults. *Epilepsy Res.* 6, 66–81.
- Gilliam, F.G., Barry, J.J., Hermann, B.P., Meador, K.J., Vahle, V., Kanner, A.M., 2006. Rapid detection of major depression in epilepsy: a multicentre study. *Lancet Neurol.* 5, 399–405.
- Hamati-Haddad, A., Abou-Khalil, B., 1998. Epilepsy diagnosis and localization in patients with antecedent childhood febrile convulsions. *Neurology* 50, 917–922.
- Hauser, W.A., Rich, S.S., Annegers, J.F., Anderson, V.E., 1990. Seizure recurrence after a 1st unprovoked seizure: an extended follow-up. *Neurology* 40, 1163–1170.
- Hesdorffer, D.C., Hauser, W.A., Annegers, J.F., Cascino, G., 2000. Major depression is a risk factor for seizures in older adults. *Ann. Neurol.* 47, 246–249.
- Hesdorffer, D.C., Hauser, W.A., Olafsson, E., Ludvigsson, P., Kjartansson, O., 2006. Depression and suicide attempt as risk factors for incident unprovoked seizures. *Ann. Neurol.* 59, 35–41.
- Jobe, P.C., 2003. Common pathogenic mechanisms between depression and epilepsy: an experimental perspective. *Epilepsy Behav.* 4, S14–S24.
- Johnson, M.R., Milne, R.L., Torn-Broers, Y., Hopper, J.L., Scheffer, I.G., Berkovic, S.F., 2003. A twin study of genetic influences on epilepsy outcome. *Twin Res.* 6, 140–146.
- Kanner, A.M., 2003. Depression in epilepsy: prevalence, clinical semiology, pathogenic mechanisms and treatment. *Biol. Psychiatry* 54, 388–398.
- Kanner, A.M., 2006. Epilepsy, suicidal behaviour, and depression: do they share common pathogenic mechanisms? *Lancet Neurol.* 5, 107–108.
- Kanner, A.M., Byrne, R., Smith, M.C., Balabanov, A.J., Frey, M., 2006. Does a life-time history of depression predict a worse

- postsurgical outcome following a left temporal lobectomy? *Ann. Neurol.* 60 (Suppl. 10), S19.
- Kjeldsen, M.J., Corey, L.A., Christensen, K., Friis, M.J., 2003. Epileptic seizures and syndromes in twins: the importance of genetic factors. *Epilepsy Res.* 55, 137–146.
- Ko, T.S., Holmes, G.L., 1999. EEG and clinical predictors of medically intractable childhood epilepsy. *Clin. Neurophysiol.* 110, 1245–1251.
- Kwan, P., Brodie, M.J., 2000. Early identification of refractory epilepsy. *N. Engl. J. Med.* 342, 314–319.
- Leschziner, G., Jorgensen, A.L., Andrew, T., Pirmohamed, M., Williamson, P.R., Marson, A.G., Coffe, A.J., Middleditch, C., Rogers, J., Bently, D.R., Chadwick, D.W., Balding, D.J., Johnson, M.R., 2006. Clinical factors and ABCB1 polymorphisms in prediction of antiepileptic drug response: a prospective cohort study. *Lancet Neurol.* 5, 668–676.
- MacDonald, B.K., Johnson, A.L., Goodridge, D.M., Cockerell, O.C., Sander, J.W., Shorvon, S.D., 2000. Factors predicting prognosis of epilepsy after presentation with seizures. *Ann. Neurol.* 48, 833–841.
- Mattson, R.H., Cramer, J.A., Collins, J.F., 1996. Prognosis for total control of complex partial and secondarily generalized tonic clonic seizures. *Neurology* 47, 68–76.
- Mendez, M.F., Cummings, J.L., Benson, D.F., 1986. Depression in epilepsy. Significance and phenomenology. *Arch. Neurol.* 43, 766–770.
- Mohanraj, R., Brodie, M.J., 2005. Outcomes in newly diagnosed localization-related epilepsies. *Seizure* 14, 318–323.
- Mohanraj, R., Brodie, M.J., 2006. Diagnosing refractory epilepsy: response to sequential treatment schedules. *Eur. J. Neurol.* 13, 277–282.
- Mohanraj, R., Brodie, M.J., 2007. Outcomes of newly diagnosed idiopathic generalized epilepsy syndromes in a non-pediatric setting. *Acta Neurol. Scand.* 115, 204–208.
- Nelson, K.B., Ellenberg, J.H., 1976. Predictors of epilepsy in children who have experienced febrile seizures. *N. Engl. J. Med.* 295, 1029–1033.
- Pisani, F., Oteri, G., Costa, C., Di Raimondo, G., Di Perri, R., 2002. Effects of psychotropic drugs on seizure threshold. *Drug Saf.* 25, 91–110.
- Sander, J.W., Hart, Y.M., Johnson, A.L., Shorvon, S.D., 1990. National general practice study of epilepsy: newly diagnosed epileptic seizures in a general population. *Lancet* 336, 1267–1271.
- Semah, F., Picot, M.C., Adam, C., Broglin, D., Arzimanoglou, A., Balin, B., Cavalcanti, D., Baulac, M., 1998. Is the underlying cause of epilepsy a major prognostic factor for recurrence? *Neurology* 51, 1256–1262.
- Sillanpää, M., 1993. Remission of seizures and predictors of intractability in long-term follow-up. *Epilepsia* 34, 930–936.
- Smith, P.E.M., McBride, A., 1999. Illicit drugs and seizures. *Seizure* 8, 441–443.
- Zagnoni, P.G., Albano, C., 2002. Psychostimulants and epilepsy. *Epilepsia* 43 (Suppl. 2), 28–31.